

Virtual Machines

Definition

A **virtual machine** is a software-created machine that behaves like real hardware. It allows one physical computer to run multiple independent operating systems or environments. Based on user text :contentReference[oaicite:0]index=0

1. Basic Idea

Main Concept

Virtualization separates two important functions:

- **multiprogramming**, and
- providing a more convenient **extended machine**.

A virtual-machine system gives several virtual machines, each behaving like a real machine.

Key Observation

Each virtual machine can act like a complete copy of the bare hardware, including:

- kernel/user mode,
- I/O,
- interrupts,
- and other hardware features.

2. VM/370

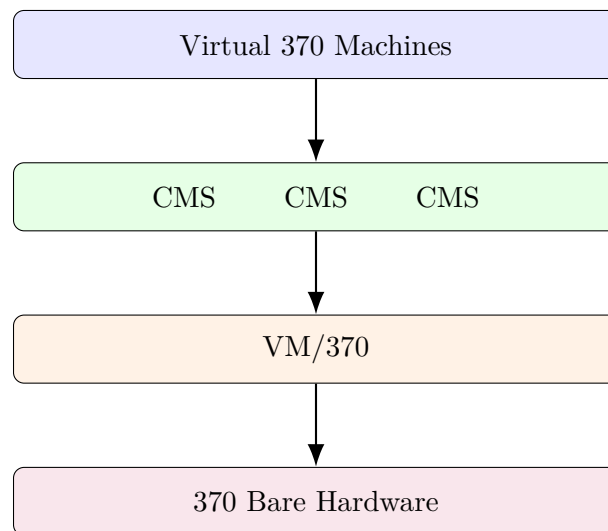
Historical Example

An important early virtual-machine system was IBM's **VM/370**, originally called **CP/CMS**. It became a major virtualization system for IBM mainframes.

Main Idea of VM/370

The heart of VM/370 was the **virtual machine monitor**, which ran directly on bare hardware and created multiple virtual machines above it.

2.1 VM/370 Structure



3. Why Virtual Machines Were Useful

Benefit

Since each virtual machine looked like a real machine, each one could run any operating system that could run on the actual hardware.

Examples

Different virtual machines could run:

- OS/360,
- transaction-processing systems,
- or CMS for interactive users.

Important Result

Different virtual machines on the same physical computer could run **different operating systems at the same time**.

4. How a System Call Worked in VM/370

Behavior

When a program running inside CMS made a system call, it was trapped to the operating system inside that **same virtual machine**, not directly to VM/370.

I/O Handling

When the guest operating system later issued hardware I/O instructions, those instructions were trapped by VM/370, which simulated the real hardware behavior.

5. Why Virtualization Became Popular Again

Rediscovery

Although virtualization existed for decades on mainframes, it became especially important again in the PC and server world because of new practical needs.

Major Reasons

Important reasons include:

- running multiple servers on one machine,
- isolating failures between services,
- reducing hosting cost,
- running multiple operating systems together,
- improving flexibility for users and companies.

6. Server Consolidation

title

Many organizations used to run mail servers, web servers, FTP servers, and other services on separate machines. Virtualization allows all of them to run on one machine while remaining separated.

Advantage

If one virtual server crashes, the others can continue running. This improves isolation and saves hardware cost.

7. Web Hosting Use

Virtual Hosting

Virtual machines are popular in web hosting because one physical machine can host many independent virtual machines.

Benefit to Customers

A customer renting a virtual machine can:

- choose their own operating system,
- install their own software,
- get more freedom than shared hosting,
- pay much less than for a dedicated server.

8. Running Multiple OS on One Computer

End-User Use

Virtualization is also useful for users who want to run more than one operating system at the same time, such as Windows and Linux on one machine.

9. Hypervisors

Hypervisor

A **hypervisor** is the software layer that creates and manages virtual machines. The older name **virtual machine monitor** is still also used.

9.1 Type 1 and Type 2 Hypervisors

Type 1 Hypervisor

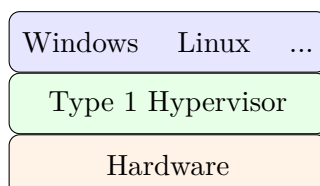
A **type 1 hypervisor** runs directly on the hardware and manages virtual machines without relying on a host operating system.

Type 2 Hypervisor

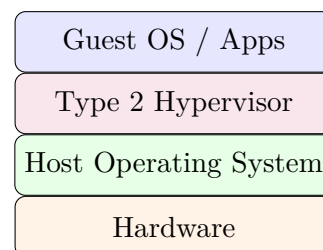
A **type 2 hypervisor** runs on top of a host operating system and uses the host system for services such as file storage and process creation.

9.2 Simple Hypervisor Comparison

(a) Type 1 Hypervisor



(b) Type 2 Hypervisor



10. Virtualizability Problem

CPU Requirement

To run virtual-machine software efficiently, the CPU should be **virtualizable**. This means privileged instructions must trap correctly so the hypervisor can handle them.

Old Problem

On some older processors, privileged instructions executed in user mode were simply ignored instead of trapping. This made efficient virtualization very difficult.

11. Machine Simulators and Binary Translation

Machine Simulator

One solution was to use a **machine simulator**, which translates and runs machine instructions in software.

Binary Translation

Some systems improved performance by translating blocks of code on the fly, storing them in a cache, and reusing them later. This technique is called **binary translation**.

Performance

Binary translation improved performance compared with simple interpreters, but it was still often not fast enough for demanding commercial use.

12. Hybrid Type 2 Hypervisors

Practical Improvement

Performance improved further when type 2 hypervisors added a kernel module to help with low-level work.

Result

Modern commercial type 2 hypervisors often use this hybrid design: part runs in user mode and part runs with kernel assistance.

13. Examples of Popular Hypervisors

Examples

Popular hypervisors include:

- VMware Workstation
- Xen
- KVM
- VirtualBox
- Hyper-V

14. Type 1 vs Type 2 Difference

Main Practical Difference

The main difference is:

- a type 2 hypervisor uses a **host operating system**,
- a type 1 hypervisor does **not** depend on a host operating system.

Storage Example

A type 2 hypervisor can store a virtual disk as a file inside the host operating system's file system.

A type 1 hypervisor must manage storage itself, usually on a raw disk partition.

15. Guest Operating System

Guest OS

The operating system running inside a virtual machine is called the **guest operating system**.

User Experience

After booting, the guest operating system behaves much like it would on real hardware, even though it is actually running inside a virtual environment.

16. Paravirtualization

Paravirtualization

A different approach is **paravirtualization**, where the operating system is modified to avoid problematic control instructions instead of simulating a completely unchanged machine.

Difference

Paravirtualization is related to virtualization, but it is not the same as true full virtualization.

17. Java Virtual Machine

JVM

Another important virtual machine is the **Java Virtual Machine (JVM)**. It is used to run Java programs in a portable way.

How It Works

- Java source code is compiled into JVM code,
- the JVM interpreter or runtime executes that code,
- the same JVM program can run on many different platforms.

Main Advantage

Because Java programs target the JVM instead of a specific hardware CPU, they can be distributed and run on many different machines more easily.

Security Advantage

If implemented properly, the JVM can check incoming programs for safety and execute them in a protected environment.

18. Advantages of Virtual Machines

Main Advantages

Virtual machines provide:

- isolation between systems,
- better hardware utilization,
- support for multiple operating systems,
- flexibility in hosting and testing,
- portability in systems like JVM.

19. Summary Points

Quick Review

Virtual machines:

- simulate complete computers,
- are managed by hypervisors,
- may run multiple guest operating systems,
- are useful in servers, hosting, and personal computing,
- also appear in other forms such as the JVM.